DAYANANDA SAGAR
College of Arts, Science & Commerce



PG – 504

II Semester M.C.A. Examination, October/November 2020
(CBCS)
COMPUTER SCIENCE
MCA202T – Database Management System

Time : 3 Hours Max. Marks : 70

SECTION – A

Answer **any five** questions. **Each** question carries **6** marks. **(5×6=30)**

1. Define DBMS and explain advantages of using DBMS.
2. What is Schema ? Explain with neat diagram 3 schema architecture of DBMS.
3. What is entity ? Explain the types of entity and attributes.
4. Write a note on Buffering of Blocks.
5. Explain domain constraint, key constraint, referential integrity constraint.
6. Describe views in SQL with an example.
7. Discuss ACID properties of the transaction.
8. Explain serial, non serial and conflict serializable schedules.

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SECTION – B

Answer **any four** questions. **Each** question carries **10** marks. **(4×10=40)**

9. Explain different types of database users.
10. Build a Entity Relationship diagram for Library Management System.
11. Explain BCNF and Normalize the following relation up to 3NF

    Empno, Ename, jobcode, jobtitle, projectcode, projecttitle, numberofhours.

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12. Using the following tables write the SQL statement to solve the questions

Customer(custid,custname,location)

Product(productid,prodname, price)

Order_header(orderid,orderdate,custid,totalamount)

Order_detail(orderid,productid,qty)

Write SQL Queries for the following

1) Find name of all customer who live in Bangalore.
2) Find the number of orders placed by each customer.
3) List order details along with custname and prodname.
4) Find total purchases made product wise.
5) List the orders which are placed in the month of May.

13. Explain Two Phase Locking Protocol used in concurrency control.

14. Write a note on :
    a) PL/SQL
    b) States of Transaction.

(2×5=10)

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II Semester M.C.A. Examination, November/December 2022
(CBCS) (2020 – 21 and Onwards)
COMPUTER SCIENCE
2MCA2 : Database Management System

Time : 3 Hours Max. Marks : 70

Instruction *: Answer* ***any five*** *from Section* ***A*** *and* ***any four*** *from Section* ***B****.*

SECTION – A

Answer **any five** questions. **Each** question carries **six** marks. **(5×6=30)**

1. Define DBMS. What are the advantages of using DBMS over traditional file system ?
2. What is Schema ? Explain with a neat diagram 3 schema architecture of DBMS.
3. Write a note on hashing techniques.
4. Explain domain constraint, key constraint and referential integrity constraints.
5. Explain views in SQL with example.
6. Discuss ACID properties of a transaction.
7. Explain relational algebra select, project and join operations with example.
8. What is scheduling a transaction ? Describe serial, non-serial and conflict serializable schedules.

SECTION – B

Answer **any four** questions. **Each** question carries **ten** marks. **(4×10=40)**

9. Explain different types of database users. **10**
10. Build a entity relationship diagram for Covid Vaccination Management System. Assume that there a 'n' hospitals 'm' doctors an 'k' citizens. Also assume that citizens have to take two doses of vaccine and there are two types of vaccines. Senior citizens have to be given priority. **10**

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11. Explain 1NF, 2NF and 3NF with comparison between 3NF and BCNF. **10**

12. Using the following tables write the SQL statement to solve below given questions. **10**

    Customer (custid, custname, location)

    Product (productid, prodname, price)

    Order-hd (orderid, orderdt, custid, totamt)

    Order-dt (orderid, productid, qty)

    Write SQL queries for following.

    1) Create above table with suitable primary key and foreign keys.
    2) Find number of orders placed by each customer.
    3) Find total quantity of purchases made order by product name.
    4) List the orders placed in the month of 'May'.
    5) List order details where total amount is more than average amount of product bought.

13. Explain two phase locking protocol used in concurrency control mechanism. **10**

14. Write a note on :

    a) Recovery techniques. **5**

    b) Triggers. **5**

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II Semester M.C.A. Examination, November/December 2023
(CBCS) (2020 – 21 and Onwards)
2MCA2 : DATABASE MANAGEMENT SYSTEMS

Time : 3 Hours Max. Marks : 70

PART – A

Note : Answer **any five**. **(5×6=30)**

1. What is three-schema architecture ? What is physical data independence and conceptual data independence ?

2. What is primary key and super key ? What is entity integrity constraint and referential integrity constraint ?

3. Given the following tables, explain the unary relational operations SELECT (σ) and PROJECT (π).

student_tb	register_no	name	gender

teacher_tb	teacher_id	name	department	phone

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4. What is a JOIN (⋈) operation ? For the given following tables give syntax for EQUI JOIN and LEFT JOIN.

employee_tb	employee_id	name	department

phone_tb	phone_id	employee_id	phone_no

NOTE : employee_id is a foreign key in phone_tb.

5. Explain terms functional dependency, fully functional dependency and transitive dependency. What is First Normal Form and Second Normal Form ?

6. For the following table, write syntax for 5 different aggregate functions.

emp_salary_tb	emp_id	department	salary	gender

7. What are insert, delete and update anomalies ? Give examples.

8. What is a view ? Write syntax to create a view. What is the advantage of a view ?

PART – B

Note : Answer **any four.** **(4×10=40)**

9. The Government of Karnataka has implemented "Anna Bhagya" scheme from July, 2023. This scheme is implemented to help people living in Karnataka who belong to Below Poverty Line (BPL) families. In this scheme, 5 kg of rice is given to all the eligible members of the family per month, and additional ₹ 170 is given (₹ 34 per kg of rice, for 5 kg) to each of the family members. The amount given to the family and the number of kgs of rice given to the family depends on the number of eligible members. The money is transferred directly to the head of the family. Assuming you are the database administrator for this scheme, identify all the attributes, entities. Write E-R diagram after identifying the type of entity and attributes.

   [Note : Additional information :

   1) The scheme is implemented by the Department of Food and Civil Supplies.
   2) BPL card of a family is unique.
   3) One may consider Aadhar number for each family member]

10. Design all the tables for the problem given in the question (9). Please make sure that the tables are in atleast 3 NF. Please also add atleast two tuples to each of the tables.

11. What is indexing ? Why is indexing important ? What is multilevel indexing ? What is B-Tree ? How is multilevel indexing is implemented using B-tree ? Assuming you are using B-tree of order 3, explain your method for the keys :

    5, 8, 3, 6, 4, 1, 0, 9, 7

12. Write SQL syntax to create a table (with an identified primary key and a foreign key), alter the table, select operation with group by and having clause and to insert and update table.

13. What is ACID property ? In concurrent transaction, explain terms interleaving, conflict and serializability. Using directed acyclic graph prove that the following transaction is conflict serializable :

    $R_2(A), W_2(A), R_3(C), W_2(B), W_3(A), W_3(C), R_1(A), R_2(B), W_1(A), W_2(B)$

14. What is database recovery ? Explain recovery technique based on deferred update and immediate update.

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II Semester M.C.A. Examination, December 2024
(CBCS) (2020 – 21 and Onwards)
COMPUTER SCIENCE
2MCA2 : Database Management Systems

Time : 3 Hours Max. Marks : 70

Instruction : *Answer* ***any five*** *from Section –* ***A*** *and* ***any four*** *from Section –* ***B****.*

SECTION – A

Answer **any five** of the following. **Each** question carries **six** marks. **(5×6=30)**

1. What are the characteristics of database approach ?
2. Briefly explain database languages.
3. Draw ER diagram for College database management system.
4. Explain multilevel indexing with example.
5. Describe relational model constraints with example.
6. What is transaction ? Explain ACID properties of transaction.
7. Discuss dynamic SQL.
8. Write a short notes on database backup and recovery from catastrophic failures.

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SECTION – B

Answer **any four** of the following. **Each** question carries **10** marks. **(4×10=40)**

9. a) What is data model ? Explain any two data models with an example. 5

   b) Define the following terms with an example : 5

   i) Entity ii) Attribute

   iii) Relationship iv) Cardinality Ratio

10. a) Explain 1NF, 2NF and 3NF with an example. 5

    b) Explain order by, having and group by clause with an example. 5

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11. a) What are joins ? Explain different types of joins. 5

b) Explain the following terms with example : 5

i) Primary key ii) Foreign key

iii) Candidate key iv) Super key

12. a) Discuss different TCL statements with syntax and example. 5

b) Explain deadlock and starvation. 5

13. a) Write the SQL queries by referring following table : 6

Field Name	Data type
EMP No.	Number
EName	Char
DOB	Date
Dept	String
Salary	Real

i) Create the table with mentioned fields

ii) Insert 5 tuples for the table

iii) Find the sum of salaries of all employees.

iv) Find the department wise count of employees

v) Display the tuples in the order of average salaries of employees.

b) Discuss different recovery techniques in DBMS. 4

14. a) Explain UNION, INTERSECTION and SET DIFFERENCE operations. 5

b) Explain different states of transaction with a neat diagram. 5

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II Semester M.C.A. Examination, November/December 2025
(CBCS) (2020 – 21 and Onwards)
COMPUTER SCIENCE
2MCA2 : Database Management System

Time : 3 Hours Max. Marks : 70

Instruction : *Answer **any five** questions from Section – **A** and **any four** questions from Section – **B**.*

SECTION – A

I. Answer **any five** of the following questions. **Each** carries **6** marks. **(5×6=30)**

1) What are the main characteristics of DBMS ?

2) Define data independence. Distinguish between physical and logical data independence.

3) Design an ER-diagram for keeping track of information about company database.

4) Discuss the ACID properties of database transaction.

5) Explain different types of JOIN in SQL.

6) Discuss Embedded SQL and Dynamic SQL.

7) Discuss two-phase locking protocol used in concurrency control.

8) Why do we require file organization ? Explain the types of file organization and mention differences between them.

SECTION – B

II. Answer **any four** of the following questions. **Each** carries **10** marks. **(4×10=40)**

9) a) What are advantages of DBMS ?

b) Explain different people behind DBMS.

10) What is normalization ? Explain briefly various types of normal forms with examples.

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